Making Effort Observable: AEIC-Mediated Mechanism Design for Multi-Agent LLM Deliberation

Tiancheng Ma[1]

School of Computer Science
author@university.edu

Abstract. Multi-agent language-model systems usually optimize the final aggregate while leaving the behavior that produced it unobserved. This is problematic when a complete, independently reasoned report is more costly than a short answer or imitation. We introduce *AEIC-MD*, a mechanism-mediated deliberation framework in which Assumption–Evidence–Inference–Conclusion (AEIC) reports form an observable interface between a mechanism designer and LLM agents. Before solving, each agent observes a public credit–cost rule and chooses among low-effort reporting, complete AEIC reasoning, verification, and peer imitation. The chosen action controls the actual generation procedure and is logged before settlement, enabling a direct follower-response test rather than retrospective response scoring.

Across 450 paired test questions from StrategyQA, MMLU-Hard, and TruthfulQA, a static AEIC process rule increases high-effort choices from 0.46% to 56.93% and majority-vote accuracy from 68.00% to 82.22% (+14.22 percentage points; Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). An outcome-only reward increases effort only to 3.08% and does not improve accuracy (67.33%, $p = 0.780$), separating observable process incentives from generic reward. Reward and cost sweeps produce the predicted monotone effort response; controlled information environments, two additional seeds, signal and history ablations, and three-to-five-agent panels delimit robustness. A held-out GLM backend does not reproduce either the effort or accuracy gain (69.33% vs. 70.67%, $p = 0.791$), establishing backend-specific contract calibration as a necessary boundary. In total, all eight preregistered suites contain 5,850 verified question-level records with independently recomputed summaries. The results support a bounded claim: public rules over verifiable report structure can change follower effort and improve collective decisions, but their semantics do not automatically transfer across model families.

Keywords: Multi-agent LLM systems · Mechanism design · Structured reasoning · Deliberation · Follower response

1 Introduction

LLM agents can debate, critique, vote, and synthesize one another's outputs. Yet the aggregate is only as useful as the behavior that produced it. A short

answer, an independently reasoned report, a verification pass, and imitation of a visible peer have different costs and informational value. Most multi-agent evaluations observe only the final answer, making these production choices invisible and leaving the system designer unable to distinguish genuine deliberation from cheap agreement.

This is a mechanism-design problem before it is an aggregation problem. A designer must announce what behavior will receive credit before the agents act; the agents must then choose whether the expected credit justifies the effort. Retrospectively weighting an already-generated response cannot establish this follower response, because the generation policy never observed the rule. Similarly, changing communication topology controls where information flows but not whether agents invest in producing independently checkable information.

We introduce **AEIC-MD**, an AEIC-mediated leader–follower framework for multi-agent deliberation. Each report exposes four fields—Assumption, Evidence, Inference, and Conclusion—that form a parseable report space. Before solving, every follower observes a public offer specifying the credit rule, action costs, information policy, and verification budget. It then declares one of four actions: low effort, complete AEIC reasoning, verification, or imitation. The declared action selects a real generation path; the resulting report, resource use, credit, cost, and reputation update are logged. This ordering turns the broad Stackelberg intuition of commitment followed by response into a falsifiable experiment.

Our confirmatory comparison pairs 450 held-out questions across StrategyQA, MMLU-Hard, and TruthfulQA under three rules. Static AEIC increases high-effort choices from 0.46% to 56.93% and final accuracy from 68.00% to 82.22%. Outcome-only reward raises effort only to 3.08% and leaves accuracy statistically unchanged. The distinction matters: the result is not that reward language generally improves answers, but that a locally legible process rule changes the action policy and the collective outcome.

We stress-test that interpretation instead of relying on one headline comparison. Reward and cost sweeps test directional response; evidence, social-only, copy-pressure, and counterevidence environments vary what agents can observe; signal and history ablations probe the implemented contract; two additional seeds and three-to-five-agent panels test stability. Finally, a held-out GLM backend does not respond to the same contract. This negative result prevents a universal claim and points to backend-specific contract calibration as the central unsolved problem.

Our contributions are:

1. an executable leader–follower protocol in which public rules precede logged action choices and action choices control real resource paths;
2. paired causal evidence that an AEIC process rule changes follower effort and improves collective accuracy beyond an outcome-only reward;
3. a fully frozen eight-suite evaluation, including dose response, information controls, ablations, robustness, and an explicit non-transfer result that defines the claim boundary.

2 Related Work

Multi-agent deliberation. Self-consistency aggregates sampled reasoning paths [13]; multi-agent debate exposes model instances to one another before aggregation [5]; and systems such as ChatEval and AgentVerse organize specialized agents or evaluators [1,3]. These approaches primarily ask how communication and aggregation affect the final answer. Recent work also documents conformity, sycophancy, and the loss of useful disagreement in homogeneous panels [9,12]. AEIC-MD addresses an earlier decision: whether an agent chooses to incur the cost of an independently checkable report before communication and aggregation occur.

Structured reasoning as an interface. Chain-of-thought, Tree-of-Thoughts, and ReAct structure an individual model's computation [15,16,17]. Our use of structure is different. AEIC partitions a report into Assumption, Evidence, Inference, and Conclusion so that report completeness, the final stance, and the resource-producing action can be observed and logged consistently across agents. The structure is therefore both a generation protocol and a contractible report space; we do not assume that formatting alone makes the content truthful.

Mechanism design for language models. Principal–agent and contract-theoretic views of LLM systems distinguish a designer's committed rule from an agent's response [10,11]. Stackelberg incentives have similarly been used in distributed learning and crowdsourcing [6,7,14]. AEIC-MD instantiates this ordering directly: the rule is rendered before generation, the follower declares an action before solving, and the corresponding resource path is executed. This differs from assigning an offline score to a response that was generated without seeing the rule.

Communication structure versus production incentives. Topology determines which reports are visible and how they flow, as in sequential, centralized, or fully connected deliberation [8,2]. It cannot by itself determine whether those reports contain costly independent work. We therefore treat information structure as a controlled environment and a future design layer, not as the identity of the mechanism. This separation is central to the experimental design: the confirmatory test holds information flow fixed while changing only the announced rule.

3 AEIC as an Observable Report Space

For question x, agent i submits

$$r_i = (A_i, E_i, I_i, C_i), \tag{1}$$

where A_i states the operative assumptions, E_i is a list of specific evidence items, I_i connects evidence to the answer, and C_i contains an explicit final stance. The

report is serialized as a strict JSON object with exactly these four fields. This design supplies three observables that free-form deliberation usually conflates: whether the report is structurally complete, what evidence is exposed to later agents, and what answer is finally adopted.

AEIC is not treated as proof of correctness. A complete report may be coherently wrong, and an agent may imitate a peer while filling every field. Instead, AEIC provides a locally checkable surface on which a mechanism can condition credit. For a report r, define completeness

$$q_{\text{comp}}(r) = \frac{1}{4} \sum_{z \in \{A,E,I,C\}} \mathbf{1}[z \text{ is non-empty}], \tag{2}$$

and let $q_C(r)$ indicate whether C yields a valid task stance. The implemented static rule uses only these process-level signals. It never receives the ground-truth answer as an input to the follower's choice prompt.

The distinction between *observable process* and *truth* is deliberate. Outcome-only credit requires the environment to know correctness after the task, whereas process credit can be announced and checked from the report itself. The comparison between these two rules asks whether a reward attached to a delayed outcome has the same behavioral effect as one attached to an immediately legible production process.

4 AEIC-Mediated Leader–Follower Mechanism

4.1 Timing and Action Space

The mechanism designer is a leader that commits to a public offer m before generation. A follower observes the offer and chooses

$$a_i \in \mathcal{A} = \{\textsc{Low}, \text{AEIC}, \textsc{Verify}, \textsc{Imitate}\}. \tag{3}$$

The four actions respectively trigger a short low-cost report, independent complete AEIC reasoning, a draft followed by an additional verification call, or a short synthesis of visible peer information. Thus, action labels are not retrospective annotations: they select different prompts, token limits, and in the tiered executor potentially different model resources.

An offer contains a visible credit rule g_m, action costs $c_m(a)$, an information policy, and a verification budget. The choice prompt explicitly asks the follower to maximize expected net utility,

$$U_i(a_i; m, h_i) = \mathbb{E}[g_m(r_i, y, h_i) \mid a_i, m, h_i] - c_m(a_i), \tag{4}$$

where h_i denotes visible history and y is available to the settlement environment only when the rule requires outcome verification. We make no claim that the language model computes Eq. (4) exactly; the observable implication is whether its action distribution changes when the revealed reward or cost changes.

4.2 Compared Rules

The confirmatory experiment compares three offers while holding questions, seeds, panel size, number of rounds, and information policy fixed.

No mechanism. No explicit credit is offered. The action costs are visible.
Outcome only. Credit is paid for a correct final stance. The report's internal structure does not affect settlement.
Static AEIC. Credit is paid for AEIC completeness and a valid explicit conclusion. The rule is fixed across agents and rounds and does not inspect the answer key.

Let $H(a) = \mathbf{1}[a \in \{\text{AEIC}, \textsc{Verify}\}]$. Our primary behavioral estimand is the paired change in $\mathbb{E}[H]$ under a revealed rule. The primary decision estimand is the paired change in question-level majority-vote correctness. Token use is a resource outcome, not a substitute for the declared action.

4.3 Settlement and Longitudinal State

After generation, the controller computes credit, subtracts the announced action cost, and updates reputation by

$$\rho_i^t = \delta \rho_i^{t-1} + g_m(r_i^t, y, h_i^t) - c_m(a_i^t), \qquad \delta = 0.9. \tag{5}$$

The static rule does not adapt its coefficients to reputation, but the complete state is retained to support history ablations and future adaptive contracts. Ground truth is withheld from settlement until the final round. Failed attempts are rolled back; retries and repair calls remain in the action trace.

5 Experimental Protocol and Audit Trail

Synchronous panels. Each question is processed by five agents for three synchronous rounds. In round t, an agent can observe only the permitted subset of round-$t-1$ reports; agents in the same round cannot observe one another. The confirmatory suite uses the independent information condition, so no peer content is visible. Controlled information suites later expose one new evidence item, only a majority stance, full peer reports with copy pressure, or stance-opposing evidence.

Figure 2 makes the implemented ordering explicit. The controller commits first, each follower declares an action before solving, the executor runs the corresponding path, and only then does settlement create the history visible in subsequent rounds. Parse or API failures follow an auditable retry or rollback branch rather than silently changing the protocol.

Choice before execution. Every agent first receives only the public contract, its visible reputation, the permitted information, and the question. It returns a single action. A second call executes the selected action. VERIFY adds a third audit-and-revision call. A formatting repair may be called only when JSON parsing fails; it is instructed to preserve the attempted conclusion and add no facts. All calls and token counts remain logged.

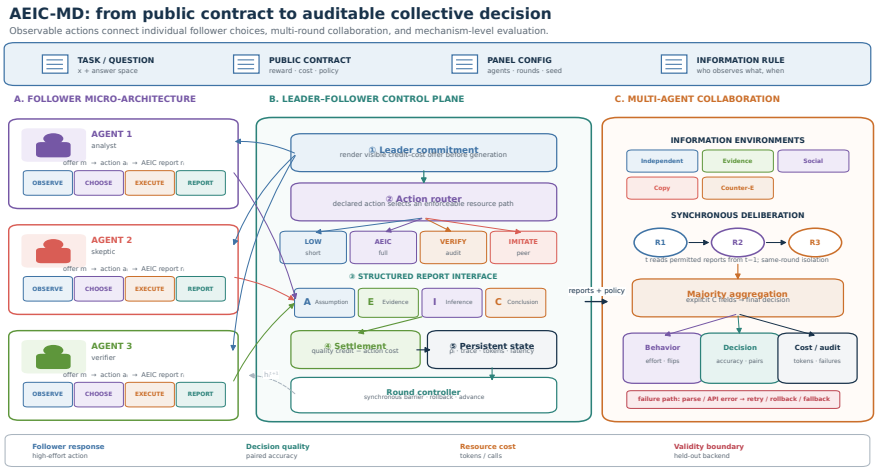


Fig. 1. AEIC-MD architecture. A public contract is rendered before followers choose resource-bearing actions. The selected path produces a structured AEIC report, after which the controller settles credit and cost, records state, and exposes only policy-permitted information to later rounds. The right-hand audit separates follower behavior, collective decision quality, and resource use.

Pairing and aggregation. All mechanisms for a question share the question-level seed. The final panel decision is majority vote over explicit C fields in the third round. High effort is defined before analysis as AEIC or VERIFY. Statistical tests operate at the natural paired unit: agent-round for the binary action and question for final correctness. We use exact McNemar tests for paired binary outcomes and paired rank tests for continuous outcomes, applying Holm correction within each registered contrast family.

Frozen provenance. Records use a versioned schema that requires the offer, chosen action, report, visible peers, reward, realized cost, reputation snapshots, raw response, action trace, latency, and token use before a record can be marked complete. Each suite passes an expected-versus-present matrix check, is frozen with SHA-256 provenance, and is independently recomputed from canonical question-level JSONL. The final manifest contains eight complete suites and 5,850 question-level records; no suite has missing, incomplete, unexpected, route-invalid, or schema-invalid records.

6 Identification and Claim Boundaries

AEIC-MD uses a Stackelberg *ordering*, not a claimed closed-form equilibrium: the designer commits to m, then followers respond. Because the LLM policy is a black box, the empirical object is the induced action distribution $\pi(a \mid x, m, h)$.

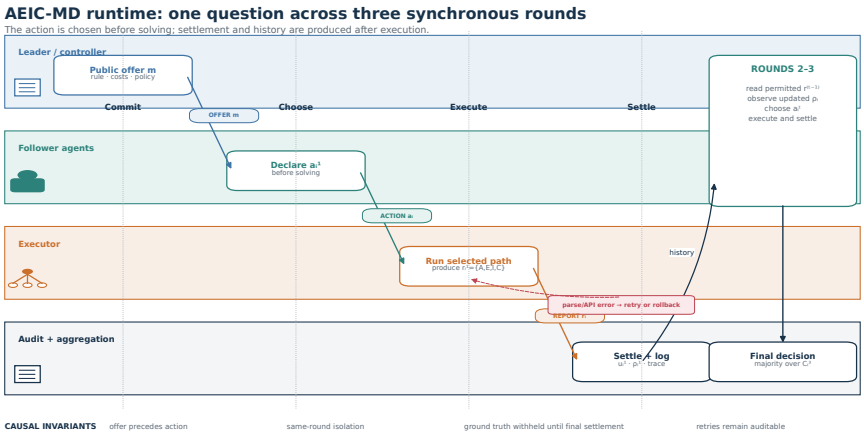


Fig. 2. Runtime sequence for one question. The four horizontal lanes separate controller commitment, follower choice, enforced execution, and settlement. Cross-round history is generated only after the current round has been logged, preserving same-round isolation.

Follower-response contrast. For paired task instances, changing m while fixing the question, seed, panel configuration, rounds, and information policy identifies the intention-to-treat effect of the announced rule on the implemented model policy. A positive difference in high-effort actions establishes rule responsiveness; it does not by itself prove utility maximization or incentive compatibility for arbitrary agents.

Outcome-only control. The contrast between Static AEIC and Outcome Only separates process observability from the mere presence of reward language. If both rules offered incentives but only the process rule changed effort and accuracy, the evidence favors the locally checkable report interface as the operative component. It does not isolate a unique contribution for each of the four AEIC fields; the registered signal ablations are descriptive bounds on the implemented contract.

Dose response. For a fixed backend and domain, Eq. (4) predicts that increasing visible reward should weakly increase high-effort choices and increasing visible effort cost should weakly decrease them. We test these directional implications without fitting a structural utility model. The boundary suite combines frozen low/high settings with a same-domain development center point, so it supports monotonicity rather than an unbiased estimate of the confirmatory treatment effect.

External validity. Prompt-mediated costs and credits have semantics supplied by the backend. Consequently, a contract that changes π for one model family need

Table 1. Confirmatory paired test across 450 questions. High effort is measured over 6,750 agent-rounds per rule; accuracy is measured over 450 questions.

Rule	High effort	Accuracy	Tokens/agent-round
No mechanism	0.46%	68.00%	663.3
Outcome only	3.08%	67.33%	689.2
Static AEIC	**56.93%**	**82.22%**	808.8

not transfer to another. We therefore treat the held-out backend as a decisive scope test and report failure to transfer in the main text. We do not claim general equilibrium existence, global leader optimality, monetary incentive compatibility, or topology-independent effectiveness.

7 Experiments

7.1 Setup

We use 150 held-out questions from each of StrategyQA, MMLU-Hard, and TruthfulQA. Each condition uses five agents, three synchronous rounds, and DeepSeek-V3 as the primary backend [4]. The confirmatory suite contains $450 \times 3 = 1{,}350$ question-condition records and 20,250 agent-rounds. Questions, seeds, panels, rounds, and aggregation are paired across No Mechanism, Outcome Only, and Static AEIC; only the visible offer changes. Final decisions use majority vote over the explicit third-round conclusions.

The two primary outcomes are high-effort choice rate—the share of AEIC and VERIFY actions—and question-level decision accuracy. We additionally report tokens per agent-round, parsing and collection integrity, and domain-level effects. Exact McNemar tests are used for paired binary outcomes and Holm-adjusted within the registered comparison family.

Figure 3 summarizes the identification logic and the status of the evidence. The confirmatory contrast changes only the public offer within paired task units. Mechanism checks and robustness suites support interpretation of that contrast, whereas the held-out backend is treated as an external-validity boundary rather than folded into a pooled positive claim.

7.2 Confirmatory Follower-Response Test

Static AEIC changes both follower behavior and the collective decision (Table 1; Figure 4). High-effort choice increases by 56.47 percentage points. Accuracy rises by 14.22 points, with 78 questions corrected and 14 harmed relative to No Mechanism (Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). The additional 145.5 tokens per agent-round are consistent with the intended resource cost of more complete work.

Outcome Only is an informative control. It raises high effort by only 2.62 points and accuracy changes by -0.67 points (27 questions corrected, 24 harmed;

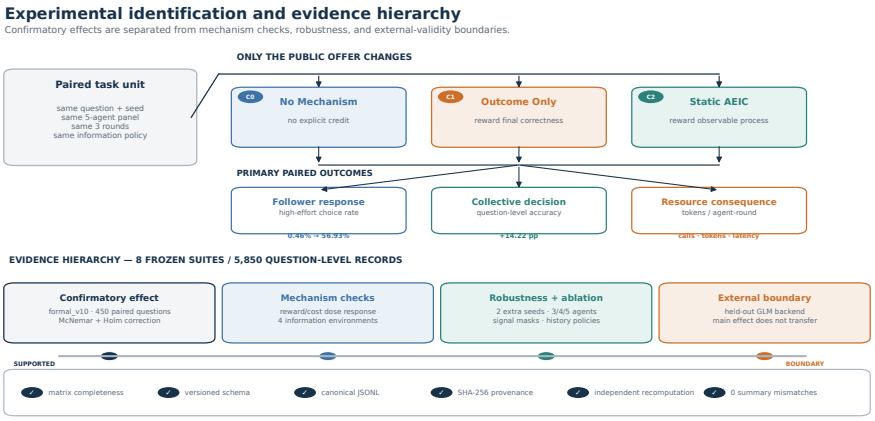


Fig. 3. Experimental identification and evidence hierarchy. Paired conditions isolate the public rule; primary outcomes measure follower response, collective accuracy, and resource use. Eight frozen suites cover confirmation, mechanism checks, robustness, ablation, and the explicit non-transfer boundary.

$p = 0.780$). Therefore, the main effect cannot be summarized as "telling the model that a reward exists." What differs is whether the rule exposes a criterion the follower can satisfy through the chosen production process.

The gain occurs in all three domains but is heterogeneous: +4.0 points on StrategyQA, +20.0 on MMLU-Hard, and +18.7 on TruthfulQA. The corresponding increases in high effort are 43.5, 73.7, and 52.2 points. This heterogeneity argues against interpreting the pooled effect as a task-independent constant.

7.3 Reward–Cost Response

For every domain, high-effort choice is monotone non-decreasing as the revealed reward scale moves from 0.05 to 0.2 to 0.5, and monotone non-increasing as the effort-cost multiplier moves from 0.5 to 1.0 to 1.5 (Figure 5). At the pooled frozen extremes, high effort rises from 49.42% at reward 0.05 to 72.84% at reward 0.5; it falls from 69.51% at cost multiplier 0.5 to 53.78% at 1.5. Because the middle points are same-domain development observations whereas the extremes are separately frozen, this is a directional mechanism check rather than a second confirmatory effect estimate.

7.4 Controlled Information Environments

We next vary only what peer information is available after the first round: one new evidence item, a majority stance without evidence, full reports plus copy pressure, or stance-opposing evidence. Across these four environments, Static

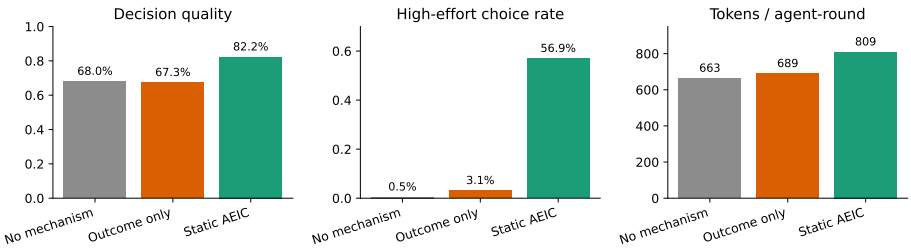


Fig. 4. Formal paired comparison. Static AEIC changes the declared effort action and improves final accuracy; an outcome-only reward does not improve accuracy.

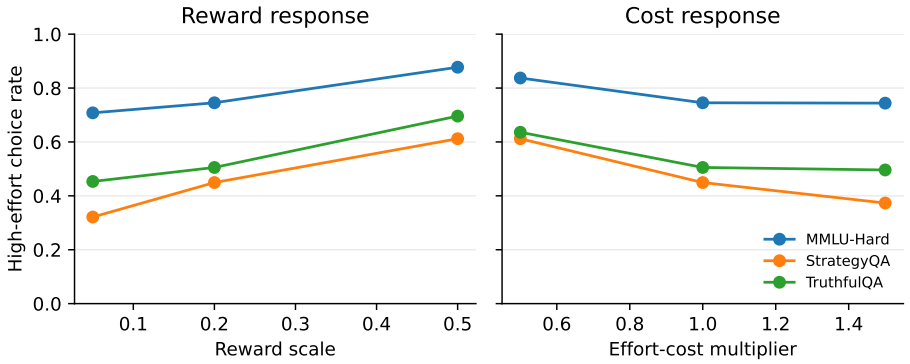


Fig. 5. Directional response to the public offer. Within every domain, effort rises with revealed reward and falls with action cost.

AEIC raises pooled high effort from 0.86% to 49.41%. Condition-specific rates under Static AEIC range from 40.89% (social only) to 59.33% (copy pressure), compared with 0.44%–1.47% without a mechanism (Figure 6). These manipulations confirm that the report interface remains behaviorally operative under distinct information structures. They do not establish that every information intervention improves accuracy or eliminates imitation.

7.5 Robustness, Ablation, and External Validity

Two additional seeds reproduce the main direction: Static AEIC reaches 85.67% accuracy versus 71.00% without a mechanism (+14.67 points; Holm-adjusted $p = 2.40 \times 10^{-9}$). With three, four, and five agents, its accuracy lies between 87.33% and 88.67%. Signal-mask and history-policy variants are summarized in Figure 7; they bound the current implementation but do not show that each AEIC field is individually necessary. In particular, removing a field can change how the backend interprets the entire visible contract, so these are intervention ablations rather than modular proofs.

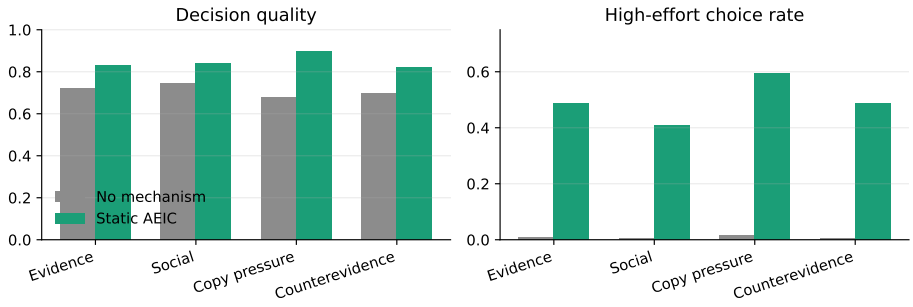


Fig. 6. High-effort actions under controlled information environments. Static AEIC shifts behavior in all four settings.

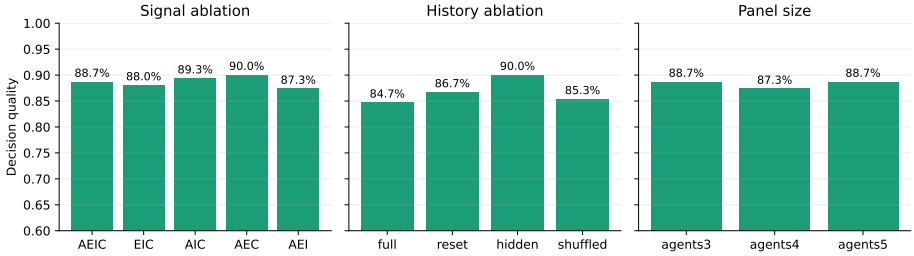


Fig. 7. Signal, history, seed, and panel-size robustness. Component variants are descriptive boundaries rather than isolated necessity proofs.

The held-out GLM backend is the strongest boundary result. Static AEIC barely changes high effort (0.58% to 0.93%, $p = 0.134$) and yields 69.33% accuracy versus 70.67% without a mechanism (Holm-adjusted $p = 0.791$). The same words and numerical schedule therefore do not define the same effective contract across model families. This failure is reported in the main text because it changes the scope of the contribution: AEIC-MD is an experimentally validated mechanism for the tested primary backend, not a universally calibrated incentive language.

7.6 Integrity and Reproducibility

Across the eight frozen suites, all 5,850 expected question-level records are present. Every suite passes schema and route validation and has an independently recomputed bundle with no summary mismatch. In the confirmatory data, API terminal failure is zero and cost logging is complete; rare parse failures are retried and remain visible in attempt logs. The manifest stores canonical hashes for each suite, preventing silent replacement of the analyzed records.

8 Discussion

Mechanism, not formatting alone. AEIC is useful here because it creates contractible observables, not because four labeled fields guarantee good reasoning. Static AEIC changes behavior when agents see a rule tied to those observables; Outcome Only shows that generic reward language is insufficient. At the same time, the signal ablations do not justify claiming that every field is independently indispensable.

Why the negative transfer matters. The GLM result indicates that prompt-mediated costs and credits are interpreted through a backend-specific policy. The contract's nominal numbers are therefore not yet a common currency. A credible next step is a calibration stage that estimates each backend's response surface on a development set before testing a fixed contract on held-out tasks. Mixed-model panels should be evaluated only after this calibration problem is addressed.

Relation to topology and deliberation. Production incentives and communication topology solve different problems. The former governs whether informative work is produced; the latter governs who can observe it and at what cost. The controlled information experiments begin to connect the two without turning topology into the paper's main causal variable. Future work can optimize information flow conditional on a calibrated reporting mechanism.

Limitations. The primary effect is established for one backend, three binary-answer QA domains, one prompt-mediated action menu, and majority aggregation. Declared actions are behavior generated by an LLM under instructions, not evidence of human-like preferences. Credits are bookkeeping quantities rather than monetary transfers, and process completeness is not factual verification. The confirmatory experiment tests a fixed static rule, not adaptive Stackelberg optimality or a general equilibrium. Finally, additional API calls and longer reports incur real compute cost even when token count is used only as a proxy.

Broader implications. The framework is most appropriate when intermediate work can be made locally inspectable and when the executor can enforce meaningful differences between actions. It should not be used to manufacture confidence from structurally complete but unverifiable reports. Logging the offer, action, execution path, and settlement is therefore part of the contribution, not merely an engineering detail.

9 Conclusion

AEIC-MD turns structured reports into an observable interface for leader–follower experimentation. A public Static-AEIC rule substantially changes the primary backend's effort choices and improves collective accuracy, whereas an outcome-only reward does not. Dose response, information controls, robustness suites, and

frozen provenance support this bounded interpretation. The held-out-backend failure is equally important: effective contracts must be calibrated to the model policy that interprets them. The resulting research direction is not ever-more elaborate topology optimization, but a disciplined sequence of observable effort, calibrated incentives, and only then adaptive information flow.

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